Akoizo Original Practice - Form 3

Division C | Protein Modeling | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Alpha-helix H bonds mainly involve…

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. Backbone groups

B. Only sulfur atoms

C. Only DNA bases

D. Only neighboring side-chain colors

Answer: _____

2. (2 points) HA contributes to…

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

A. Attachment and fusion

B. Only genome replication

C. Only salt sensing

D. Only ATP storage

Answer: _____

3. (2 points) An unsupported missing loop should be…

Unit: Event evidence, reference design, and model purpose.
Worked-case data: A structure entry has chains A and B and a gap at residues 40-45. Plan a model label.

A. Presented as directly observed

B. Deleted from the sequence silently

C. Called an active site

D. Marked as missing or modeled

Answer: _____

4. (2 points) A disulfide links…

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

A. Any two glycines

B. DNA bases

C. Two phosphate groups

D. Cysteines

Answer: _____

5. (2 points) Color by structure shows…

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain's helix region.

A. Measured free energy

B. Every atom's charge

C. A diagnosis

D. A structural classification

Answer: _____

6. (2 points) A protein may require…

Unit: Protein synthesis and diverse functions.
Worked-case data: Coding DNA is 5′-ATG GAA TAA-3′. Give simple unprocessed mRNA and translated product.

A. No folding

B. No chemical environment

C. Partners or modifications

D. Only its name

Answer: _____

7. (2 points) Binding alone proves neutralization?

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Only for any antibody

B. Only in a diagram

C. No

D. Always

Answer: _____

8. (2 points) Model review should use…

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

A. Only appearance

B. An unrelated year automatically

C. The verified applicable rubric

D. Invented point values

Answer: _____

9. (4 points) Compare motif and domain.

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

Answer: _____
_____
_____

10. (4 points) Why verify residue numbering?

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

Answer: _____
_____
_____

11. (4 points) What should an interaction annotation include?

Unit: Event evidence, reference design, and model purpose.
Worked-case data: A structure entry has chains A and B and a gap at residues 40-45. Plan a model label.

Answer: _____
_____
_____

12. (4 points) Contrast rate and stability.

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

Answer: _____
_____
_____

13. (4 points) Why is projected overlap insufficient?

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain's helix region.

Answer: _____
_____
_____

14. (4 points) Why can mature numbering differ?

Unit: Protein synthesis and diverse functions.
Worked-case data: Coding DNA is 5′-ATG GAA TAA-3′. Give simple unprocessed mRNA and translated product.

Answer: _____
_____
_____

15. (4 points) What can mutation change?

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

Answer: _____
_____
_____

16. (4 points) Why use text with color?

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

Answer: _____
_____
_____

17. (4 points) What does a cartoon hide?

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

Answer: _____
_____
_____

18. (4 points) What does a prefusion structure omit?

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

Answer: _____
_____
_____